

INNOVEX

SIH 2026 - 3-Person Professional Presentation Speaking Guide

Project in PPT: AI-Based Urban Waterlogging Risk Prediction and Preventive Response System

Use this PDF for rehearsal: it assigns exactly who speaks on each slide, gives a ready-to-speak English script, short Thanglish meaning for understanding, handover lines, and safe answers for likely judge questions.

Member	Slides	Responsibility
Person 1	1 & 2	Problem + Proposed Solution + Innovation
Person 2	3 & 4	Technical Approach + Methodology + Feasibility
Person 3	5 & 6	Impact + Benefits + References + Conclusion

How to present as a 3-person team

Person 1: What is the problem and what is our solution?

Person 2: How does the system technically work, and is it feasible?

Person 3: Who benefits, what is the impact, and what are our references?

Recommended delivery: Speak naturally, do not read every bullet on the slide, and use the slide as a visual guide. Each speaker should end with the handover line so the presentation feels continuous.

PERSON 1 - SLIDE 1

TITLE PAGE - Project identity and one-sentence purpose

READY-TO-SPEAK SCRIPT

Good morning everyone. We are Team INNOVEX.

Our project is titled **AI-Based Urban Waterlogging Risk Prediction and Preventive Response System**.

Our problem statement ID is **SIH26206**, and the theme is **Disaster Management**.

The main aim of our project is to identify urban areas that have a higher risk of waterlogging and support authorities in taking preventive action.

Our approach combines rainfall information, historical waterlogging incidents and geospatial data to generate useful risk information for urban areas.

THANGLISH MEANING

Namma project oda main aim, urban area-la waterlogging risk adhigama irukkura places-ah identify panni, authorities preventive action edukka support panradhu. Rainfall, historical incidents and geospatial data combine pannuvom.

DO NOT OVER-EXPLAIN

Slide 1-la technologies, model names, map details ellam explain panna vendam. Project enna nu clear-ah introduce pannina podhum.

HANDOVER

Now I will explain our proposed solution, how it addresses the problem, and what is unique about our approach.

PERSON 1 - SLIDE 2

IDEA TITLE - Problem -> solution -> innovation

READY-TO-SPEAK SCRIPT

Urban waterlogging can affect roads, vulnerable locations, people and important facilities during heavy rainfall.

One important challenge is deciding which areas need preventive attention first, especially when manpower and budget are limited.

Our proposed system predicts waterlogging risk for **24, 48 and 72-hour horizons**.

It combines **rainfall data, historical incidents and geospatial data**. The system then shows predicted risk together with population and facility exposure, and recommends priority preventive actions.

This helps identify potentially high-risk areas early, understand who and what may be affected, prioritize drains, roads and vulnerable areas, and support decisions with limited resources.

INNOVATION - READY TO SAY

The uniqueness of our solution is the workflow:

Predict -> Explain -> Prioritize -> Optimize -> Simulate.

Predict means estimating risk. Explain means showing the evidence behind the risk. Prioritize means considering impact. Optimize means planning with available crews and budget. Simulate means testing what-if scenarios.

THANGLISH MEANING

First risk predict pannuvom. Appuram why that area risk-la irukku-nu evidence kaamippom. Affected people/facilities based on priority decide pannuvom. Limited crew and budget-ku best action select pannuvom. Different rainfall/resource scenarios-ai simulation panni compare pannuvom.

IMPORTANT SAFETY

Do not say: "100% waterlogging will happen." Say: "The system estimates waterlogging risk." A risk prediction is not a guarantee.

HANDOVER

Now I will hand over to my teammate to explain the technical approach and how this workflow is implemented.

PERSON 2 - SLIDE 3

TECHNICAL APPROACH - Technology stack and end-to-end methodology

READY-TO-SPEAK SCRIPT

Now I will explain the technical approach of our system.

For the frontend, we use **Next.js, React, TypeScript and Tailwind CSS**.

For the backend, we use **FastAPI and Python**.

For the database, we use **PostgreSQL with PostGIS**.

For geospatial work, we use **Leaflet and GeoPandas**. For data processing, we use **Pandas and NumPy**.

Machine Learning is used for risk prediction, **SHAP** is used for explainability, and **Google OR-Tools** is used for optimization. Git and GitHub are used for version control.

METHODOLOGY - POINT TO THE FLOW

The process starts with **rainfall, historical incidents and GIS data**.

First, the data is processed and cleaned. Next, we perform geospatial analysis.

Then the system performs risk prediction. After that, we analyse impact and exposure.

Based on those results, a priority score is calculated. We then optimize crew and budget allocation.

We can also run what-if simulations, and the final results are shown through a municipal dashboard.

THANGLISH MEANING

Data collect pannitu clean pannuvom. GIS analysis-la terrain/drainage/location relationships paapom. ML risk estimate pannum. Exposure and priority calculate pannuvom. OR-Tools resource allocation optimize pannum. Final result dashboard-la show aagum.

JUDGE QUESTION - HOW CAN YOU PREDICT BEFORE RAINFALL?

When forecast rainfall information is available, it can be combined with historical incidents and geographical factors to estimate risk for different areas. We present this as a risk estimate, not a certainty.

HANDOVER

Next, I will explain the feasibility, key challenges and how we plan to mitigate them.

PERSON 2 - SLIDE 4

FEASIBILITY AND VIABILITY - What is feasible, what can go wrong, and how we handle it

FEASIBILITY - READY TO SAY

Our idea is technically feasible because it can use rainfall, historical incident and geospatial datasets.

The web-based architecture allows modular development. GIS, Machine Learning and optimization can be integrated into one workflow.

The system can also be scaled from a demonstration ward to larger urban areas.

CHALLENGES & RISKS - READY TO SAY

There are also practical challenges.

Local waterlogging data may be limited or inconsistent. Drainage and infrastructure data may be incomplete. Prediction uncertainty can occur because of data quality.

Also, different cities may require different datasets and risk factors.

MITIGATION - READY TO SAY

To address these challenges, we validate and clean datasets before model use.

We combine historical, spatial and rainfall features and display supporting evidence with risk results.

Models can be retrained and calibrated when new data becomes available. We also clearly distinguish real, public and synthetic or demonstration data.

THANGLISH MEANING

Main problem technology illa; data availability and quality. Local waterlogging/drain data incomplete irukkalam. So validate/clean panni, multiple data sources combine panni, new data vandha model improve pannanum.

HANDOVER

Now I will hand over to my teammate to explain the target users, impact and benefits.

PERSON 3 - SLIDE 5

IMPACT AND BENEFITS - Who uses it and what value it creates

READY-TO-SPEAK SCRIPT

The primary target users of our system are **municipal authorities, disaster management teams, field response crews, ward engineers and citizens.**

The first impact is early identification of potential waterlogging risk.

Second, it supports better prioritization of vulnerable areas.

Third, it provides visibility of population and critical-facility exposure.

Finally, it can support more efficient use of limited crews and budgets.

EXAMPLE - SAY THIS

For example, when an area is identified as a higher-risk area, authorities can use the available information to prioritize that location for preventive inspection or intervention before the situation becomes severe.

BENEFITS - READY TO SAY

The solution supports **proactive rather than reactive action.**

It can help reduce avoidable disruption during heavy rainfall, enable data-driven municipal decisions, and provide a scalable framework for urban waterlogging management.

THANGLISH MEANING

Use panra people municipality, disaster teams, field crews, ward engineers. System early risk identify panni, vulnerable area priority decide panna help pannum. Limited crew/budget-a better use panna support pannum.

IMPORTANT

Avoid claiming that the system guarantees a flood or a specific percentage of damage reduction unless you have measured evidence.

PERSON 3 - SLIDE 6

RESEARCH AND REFERENCES + CONCLUSION - Credible sources and final message

REFERENCES - READY TO SAY

For our research and data references, we considered several relevant sources.

The first is the **India Meteorological Department, IMD**, relevant for rainfall and weather data.

The second is the **India Open Government Data Platform**, which provides a rainfall data catalogue.

The third is the **National Remote Sensing Centre and NDEM**, relevant for flood hazard and hydrological disaster data.

We also use **OpenStreetMap** as a geospatial reference for roads, waterways, amenities and other map information.

CONCLUSION - READY TO SAY

To conclude, our project aims to move urban waterlogging management from reactive response towards proactive prevention.

By combining rainfall, historical incident and geospatial information, the system can estimate risk, identify potentially vulnerable areas, prioritize preventive actions and support better resource allocation.

Our goal is to provide a practical decision-support system for urban authorities.

Thank you. We are happy to answer your questions.

THANGLISH MEANING

Final message: waterlogging nadandha apram response edukkaradhukku badhil, munnaadiye risk identify panni preventive action plan panna support panradhu dhaan project.

DELIVERY TIP

Conclusion slow-ah pesunga. “Proactive prevention” and “decision-support” words clear-ah pronounce pannunga.

JUDGE Q&A; - SAFE, DIRECT ANSWERS

What is the main problem?

During heavy rainfall, authorities may not know early enough which urban areas need preventive attention first. Our system estimates waterlogging risk and supports prioritization.

How do you predict before rainfall?

When forecast rainfall is available, it is combined with historical incidents and geospatial factors to estimate risk. It is a prediction, not a guarantee.

What data do you use?

Rainfall, historical incidents and GIS/geospatial data. The solution also analyses population and critical-facility exposure.

Why 24/48/72 hours?

These are the prediction horizons specified in our proposed solution.

How do you decide priority?

The workflow considers risk, impact/exposure and the availability of limited resources. The output is a priority for preventive action.

What if government data is incomplete?

We validate and clean the available data, combine multiple data sources, show supporting evidence and retrain/calibrate as better data becomes available.

What makes your project unique?

It is not only risk prediction. The complete workflow is Predict -> Explain -> Prioritize -> Optimize -> Simulate.

What does SHAP do?

SHAP is used to support explainability by showing which features contribute to a model prediction, when the model and SHAP pipeline are connected.

What does OR-Tools do?

It is used for crew and budget optimization, so limited resources can be allocated across prioritized interventions.

Is the system live?

Only features that are actually connected should be described as live. Historical or demo datasets must be clearly labelled.

Will every high-risk area definitely flood?

No. The system provides a risk estimate based on available data; it does not guarantee flooding.

Accuracy rule for the whole team: Do not invent accuracy percentages, real-time government connections, population numbers, cost savings, or flood certainty. Describe only what is actually implemented or clearly label something as demo/simulation.

30-MINUTE REHEARSAL CHECKLIST

1. Person 1 rehearses Slides 1-2 without reading directly from the slide.
2. Person 2 rehearses the Slide 3 flow by pointing to each stage.
3. Person 2 rehearses the data-quality answer on Slide 4.
4. Person 3 rehearses the practical example on Slide 5.
5. Person 3 rehearses the conclusion on Slide 6 slowly and clearly.
6. All three practise the handover lines once.

7. Run the Q&A;; each member should be able to answer the basic questions even if another member is asked first.